

Microsoft Excel for this analysis.

Scenario

Anmol Emporium stores want to create an annual sales report for 2022. So that, they can understand their customers and grow more sales in 2023.

Anmol Emporium Stores Dataset

Question/ Business Task for report

- Compare the sales and orders using a single chart.
- Which month got the highest sales and orders?
- Who purchased more — men or women in 2022?
- What is the different order status in 2022?
- List the top 5 states that contribute to the sales.
- Relation between age and gender based on a number of orders.
- Which channel is contributing to maximum sales?
- Highest selling category? etc.

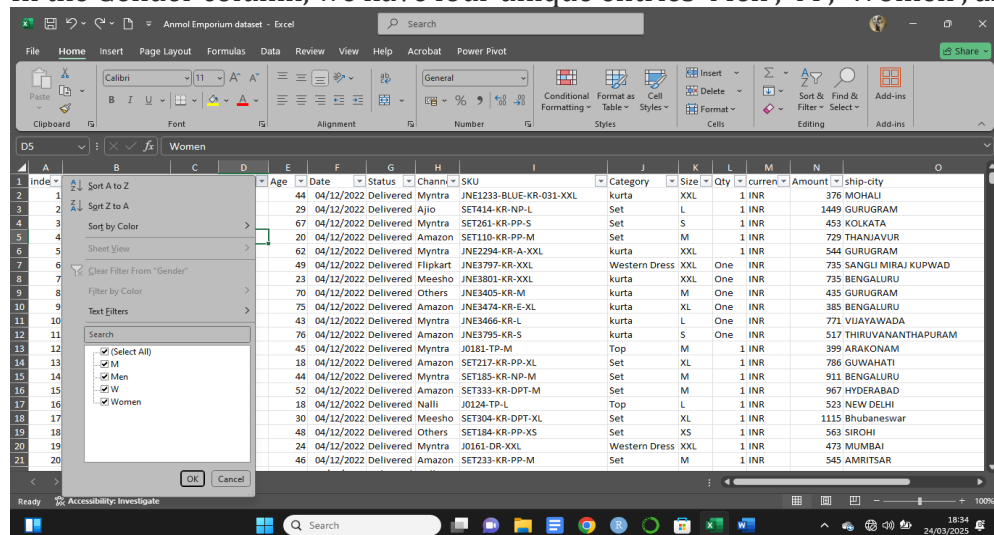
Process

First, we study the dataset, it has 19 columns and 31047 rows. I have seen it will solve our business problems and business points.

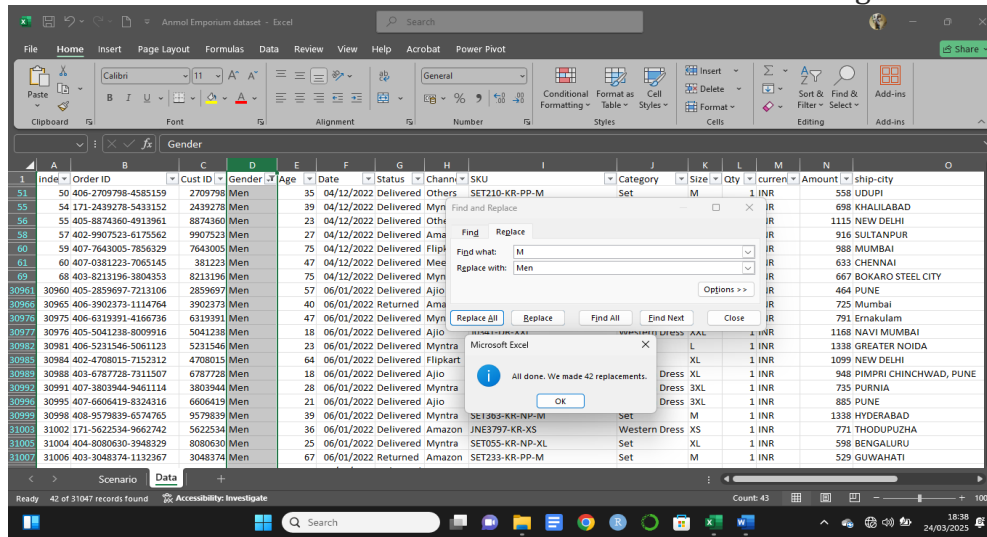
Data Cleaning

First, we have to cross-check each column that the data is correct or not.

In the Gender column, we have four unique entries 'Men', 'M', 'Women', and 'W'.

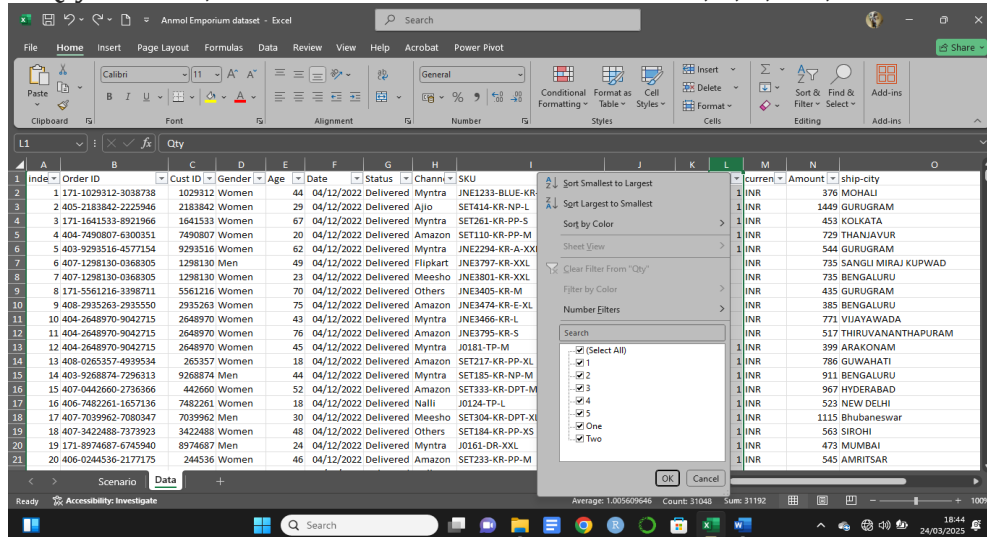


We have to convert all the 'M' to 'Men' and 'W' to 'Women' using Find and Replace in the home tab.

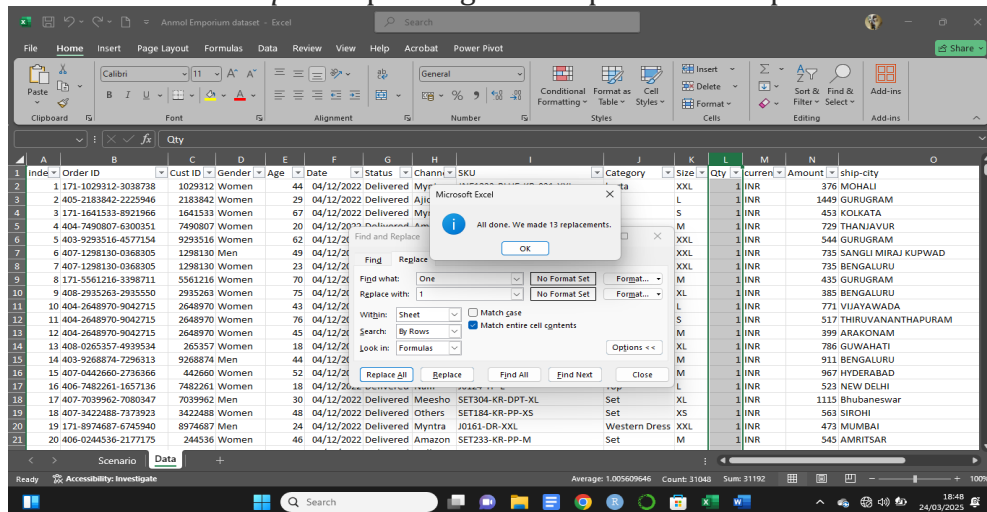


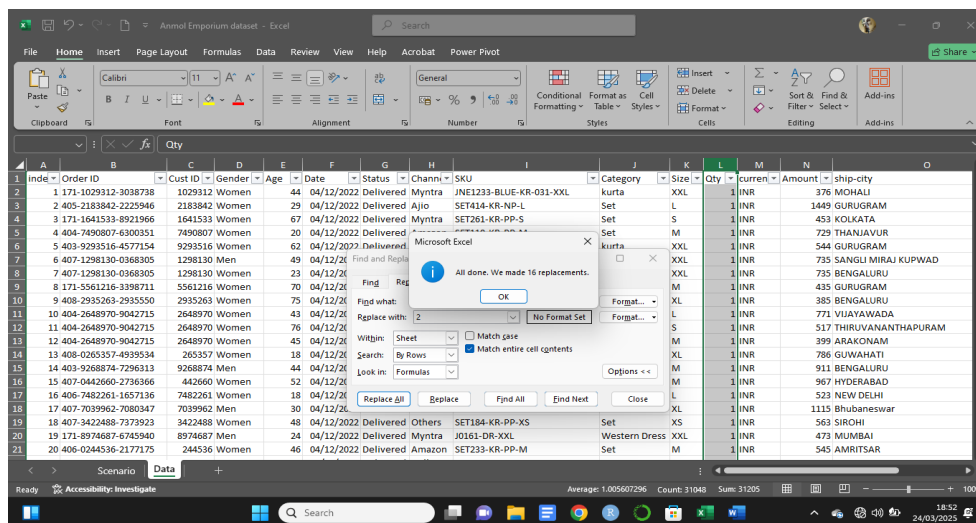
Cleaning the gender column

In Qty. column, we have some numeric values like 1, 2, 3, etc., and some alphabetic values like one and two.



Use the *Find and Replace* option again to replace these alphabetical values with numeric ones.





Replace the alphabetical values with the numeric one

Here, we complete the data-cleaning process of this dataset.

Data Processing

In this step, we removed unwanted columns then, calculating some data which help us in the analysis.

One of the business tasks is to show the relationship between age and gender.

So, we create an age group column where we divide the age into three categories:

- 50 - Above — Above 50
- 30 - 49 — between 30 and 49
- 18 - 29 — below 30

We used the nested if function to categorize the age group:

=IF(E2 >= 50, "50-Above", IF(E2 >= "30-49", "18-29"))

Excel screenshot showing the initial data table with columns: Index, Order ID, Cust ID, Gender, Age, Age Group, Date, Status, Channel, Category, Size, Qty, Amount, Ship-state, B2B. The formula bar shows: `=IF(E2>=50,"50-Above",IF(E2>=30,"30-49","18-29"))`.

Index	Order ID	Cust ID	Gender	Age	Age Group	Date	Status	Channel	Category	Size	Qty	Amount	Ship-state	B2B
1	171-1029312-3038738	1029312	Women	44	30-49	04/12/2022	Delivered	Myntra	kurta	XXL	1	376	PUNJAB	FALSE
2	405-2183842-2225946	2183842	Women	29	18-29	04/12/2022	Delivered	Ajio	Set	L	1	1449	HARYANA	FALSE
3	171-1641533-8921966	1641533	Women	67	50-Above	04/12/2022	Delivered	Myntra	Set	S	1	453	WEST BENGAL	FALSE
4	404-7490807-6300351	7490807	Women	20	18-29	04/12/2022	Delivered	Amazon	Set	M	1	729	TAMIL NADU	FALSE
5	503-9293516-4577154	9293516	Women	62	50-Above	04/12/2022	Delivered	Myntra	kurta	XXL	1	544	HARYANA	FALSE
6	407-1298130-0368305	1298130	Men	49	30-49	04/12/2022	Delivered	Flipkart	Western Dress	XXL	1	735	MAHARASHTRA	FALSE
7	407-1298130-0368305	1298130	Women	23	18-29	04/12/2022	Delivered	Meesho	kurta	XXL	1	735	KARNATAKA	FALSE
8	171-5561216-3398711	5561216	Women	70	50-Above	04/12/2022	Delivered	Others	kurta	M	1	435	HARYANA	FALSE
9	408-2935263-2935500	2935263	Women	75	50-Above	04/12/2022	Delivered	Amazon	kurta	XL	1	385	KARNATAKA	FALSE
10	104-2648970-9042715	2648970	Women	43	30-49	04/12/2022	Delivered	Myntra	kurta	L	1	771	ANDHRA PRADE	FALSE
11	104-2648970-9042715	2648970	Women	76	50-Above	04/12/2022	Delivered	Amazon	kurta	S	1	517	KERALA	FALSE
12	104-2648970-9042715	2648970	Women	45	30-49	04/12/2022	Delivered	Myntra	Top	M	1	399	TAMIL NADU	FALSE
13	408-0265357-4939534	265357	Women	18	18-29	04/12/2022	Delivered	Amazon	Set	XL	1	786	ASSAM	FALSE
14	403-9268874-7296313	9268874	Men	44	30-49	04/12/2022	Delivered	Myntra	Set	M	1	911	KARNATAKA	FALSE
15	1507-0442660-2736366	442660	Women	52	50-Above	04/12/2022	Delivered	Amazon	Set	M	1	967	TELANGANA	FALSE
16	406-7482261-1657136	7482261	Women	18	18-29	04/12/2022	Delivered	Nalli	Top	L	1	523	DELHI	FALSE
17	407-7039962-7080347	7039962	Men	30	30-49	04/12/2022	Delivered	Meesho	Set	XL	1	1115	ODISHA	FALSE
18	407-3422488-7373923	3422488	Women	48	30-49	04/12/2022	Delivered	Others	Set	XS	1	563	RAJASTHAN	FALSE
19	171-8974687-6745940	8974687	Men	24	18-29	04/12/2022	Delivered	Myntra	Western Dress	XXL	1	473	MAHARASHTRA	FALSE
20	406-0244536-2177175	244536	Women	46	30-49	04/12/2022	Delivered	Amazon	Set	M	1	545	PUNJAB	FALSE

Excel screenshot showing the data table with the 'Age Group' column highlighted. The formula bar shows: `=IF(E2>=50,"50-Above",IF(E2>=30,"30-49","18-29"))`.

Index	Order ID	Cust ID	Gender	Age	Age Group	Date	Status	Channel	Category	Size	Qty	Amount	Ship-state	B2B
1	171-1029312-3038738	1029312	Women	44	30-49	04/12/2022	Delivered	Myntra	kurta	XXL	1	376	PUNJAB	FALSE
2	405-2183842-2225946	2183842	Women	29	18-29	04/12/2022	Delivered	Ajio	Set	L	1	1449	HARYANA	FALSE
3	171-1641533-8921966	1641533	Women	67	50-Above	04/12/2022	Delivered	Myntra	Set	S	1	453	WEST BENGAL	FALSE
4	404-7490807-6300351	7490807	Women	20	18-29	04/12/2022	Delivered	Amazon	Set	M	1	729	TAMIL NADU	FALSE
5	503-9293516-4577154	9293516	Women	62	50-Above	04/12/2022	Delivered	Myntra	kurta	XXL	1	544	HARYANA	FALSE
6	407-1298130-0368305	1298130	Men	49	30-49	04/12/2022	Delivered	Flipkart	Western Dress	XXL	1	735	MAHARASHTRA	FALSE
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8	171-5561216-3398711	5561216	Women	70	50-Above	04/12/2022	Delivered	Others	kurta	M	1	435	HARYANA	FALSE
9	408-2935263-2935500	2935263	Women	75	50-Above	04/12/2022	Delivered	Amazon	kurta	XL	1	385	KARNATAKA	FALSE
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12	104-2648970-9042715	2648970	Women	45	30-49	04/12/2022	Delivered	Myntra	Top	M	1	399	TAMIL NADU	FALSE
13	408-0265357-4939534	265357	Women	18	18-29	04/12/2022	Delivered	Amazon	Set	XL	1	786	ASSAM	FALSE
14	403-9268874-7296313	9268874	Men	44	30-49	04/12/2022	Delivered	Myntra	Set	M	1	911	KARNATAKA	FALSE
15	1507-0442660-2736366	442660	Women	52	50-Above	04/12/2022	Delivered	Amazon	Set	M	1	967	TELANGANA	FALSE
16	406-7482261-1657136	7482261	Women	18	18-29	04/12/2022	Delivered	Nalli	Top	L	1	523	DELHI	FALSE
17	407-7039962-7080347	7039962	Men	30	30-49	04/12/2022	Delivered	Meesho	Set	XL	1	1115	ODISHA	FALSE
18	407-3422488-7373923	3422488	Women	48	30-49	04/12/2022	Delivered	Others	Set	XS	1	563	RAJASTHAN	FALSE
19	171-8974687-6745940	8974687	Men	24	18-29	04/12/2022	Delivered	Myntra	Western Dress	XXL	1	473	MAHARASHTRA	FALSE
20	406-0244536-2177175	244536	Women	46	30-49	04/12/2022	Delivered	Amazon	Set	M	1	545	PUNJAB	FALSE

Create an age group column using the 'IF function'

Another business task is to show the highest sales and orders month-wise.

Now, we create the month column where we process the month from the date column in text format.

Here, we use the 'Text function' to extract the month from the date column in text format.

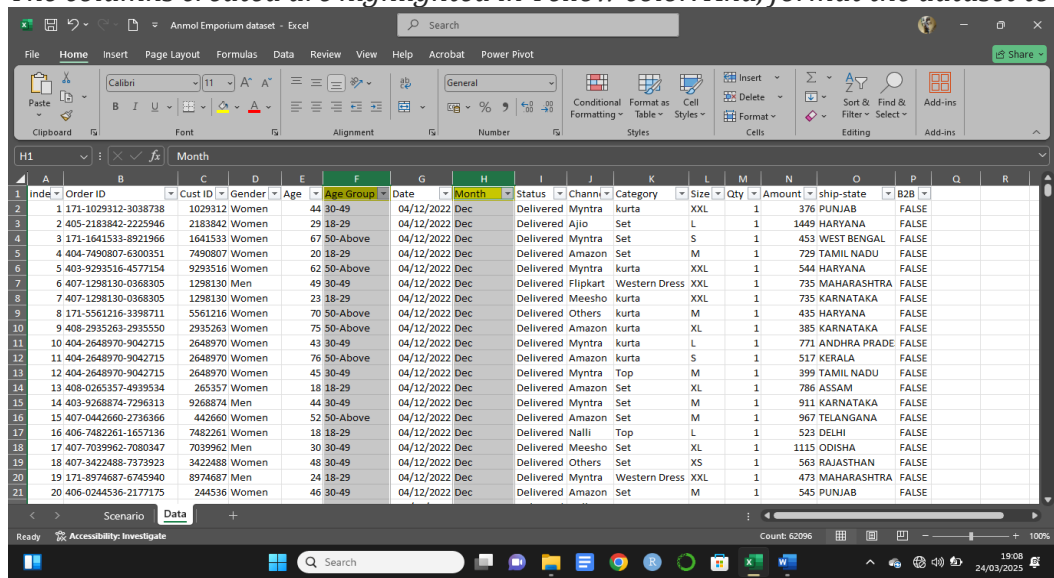
=TEXT(G2,"mmm")

Excel screenshot showing the data table with the 'Month' column added. The formula bar shows: `=TEXT(G2,"MMM")`.

Index	Order ID	Cust ID	Gender	Age	Age Group	Date	Month	Status	Channel	Category	Size	Qty	Amount	Ship-state	B2B
1	171-1029312-3038738	1029312	Women	44	30-49	04/12/2022	Dec	Delivered	Myntra	kurta	XXL	1	376	PUNJAB	FALSE
2	405-2183842-2225946	2183842	Women	29	18-29	04/12/2022	Dec	Delivered	Ajio	Set	L	1	1449	HARYANA	FALSE
3	171-1641533-8921966	1641533	Women	67	50-Above	04/12/2022	Dec	Delivered	Myntra	Set	S	1	453	WEST BENGAL	FALSE
4	404-7490807-6300351	7490807	Women	20	18-29	04/12/2022	Dec	Delivered	Amazon	Set	M	1	729	TAMIL NADU	FALSE
5	503-9293516-4577154	9293516	Women	62	50-Above	04/12/2022	Dec	Delivered	Myntra	kurta	XXL	1	544	HARYANA	FALSE
6	407-1298130-0368305	1298130	Men	49	30-49	04/12/2022	Dec	Delivered	Flipkart	Western Dress	XXL	1	735	MAHARASHTRA	FALSE
7	407-1298130-0368305	1298130	Women	23	18-29	04/12/2022	Dec	Delivered	Meesho	kurta	XXL	1	735	KARNATAKA	FALSE
8	171-5561216-3398711	5561216	Women	70	50-Above	04/12/2022	Dec	Delivered	Others	kurta	M	1	435	HARYANA	FALSE
9	408-2935263-2935500	2935263	Women	75	50-Above	04/12/2022	Dec	Delivered	Amazon	kurta	XL	1	385	KARNATAKA	FALSE
10	104-2648970-9042715	2648970	Women	43	30-49	04/12/2022	Dec	Delivered	Myntra	kurta	L	1	771	ANDHRA PRADE	FALSE
11	104-2648970-9042715	2648970	Women	76	50-Above	04/12/2022	Dec	Delivered	Amazon	kurta	S	1	517	KERALA	FALSE
12	104-2648970-9042715	2648970	Women	45	30-49	04/12/2022	Dec	Delivered	Myntra	Top	M	1	399	TAMIL NADU	FALSE
13	408-0265357-4939534	265357	Women	18	18-29	04/12/2022	Dec	Delivered	Amazon	Set	XL	1	786	ASSAM	FALSE
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15	1507-0442660-2736366	442660	Women	52	50-Above	04/12/2022	Dec	Delivered	Amazon	Set	M	1	967	TELANGANA	FALSE
16	406-7482261-1657136	7482261	Women	18	18-29	04/12/2022	Dec	Delivered	Nalli	Top	L	1	523	DELHI	FALSE
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20	406-0244536-2177175	244536	Women	46	30-49	04/12/2022	Dec	Delivered	Amazon	Set	M	1	545	PUNJAB	FALSE

Create the month column using the ‘TEXT function’

The columns created are highlighted in Yellow color. And, format the dataset to table.



Index	Order ID	Cust ID	Gender	Age	Age Group	Date	Month	Status	Channel	Category	Size	Qty	Amount	Ship State	B2B
1	171-1029312-3038738	1029312	Women	44	30-49	04/12/2022	Dec	Delivered	Myntra	kurta	XXL	1	376	PUNJAB	FALSE
2	405-2183842-2225946	2183842	Women	29	18-29	04/12/2022	Dec	Delivered	Ajio	Set	L	1	1449	HARYANA	FALSE
3	171-1641533-8921966	1641533	Women	67	50-Above	04/12/2022	Dec	Delivered	Myntra	Set	S	1	453	WEST BENGAL	FALSE
4	404-7490807-6300351	7490807	Women	20	18-29	04/12/2022	Dec	Delivered	Amazon	Set	M	1	729	TAMIL NADU	FALSE
5	403-9293516-4577154	9293516	Women	62	50-Above	04/12/2022	Dec	Delivered	Myntra	kurta	XXL	1	544	HARYANA	FALSE
6	707-1298130-0368305	1298130	Men	49	30-49	04/12/2022	Dec	Delivered	Flipkart	Western Dress	XXL	1	735	MAHARASHTRA	FALSE
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14	403-9268874-7296313	9268874	Men	44	30-49	04/12/2022	Dec	Delivered	Myntra	Set	M	1	911	KARNATAKA	FALSE
15	407-0442660-2736366	442660	Women	52	50-Above	04/12/2022	Dec	Delivered	Amazon	Set	M	1	967	TELANGANA	FALSE
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17	407-7039962-7080347	7039962	Men	30	30-49	04/12/2022	Dec	Delivered	Myntra	Set	XL	1	1115	ODISHA	FALSE
18	407-3422488-7373923	3422488	Women	48	30-49	04/12/2022	Dec	Delivered	Others	Set	XS	1	563	RAJASTHAN	FALSE
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20	406-0244536-2177175	244536	Women	46	30-49	04/12/2022	Dec	Delivered	Amazon	Set	M	1	545	PUNJAB	FALSE

Analysis stage

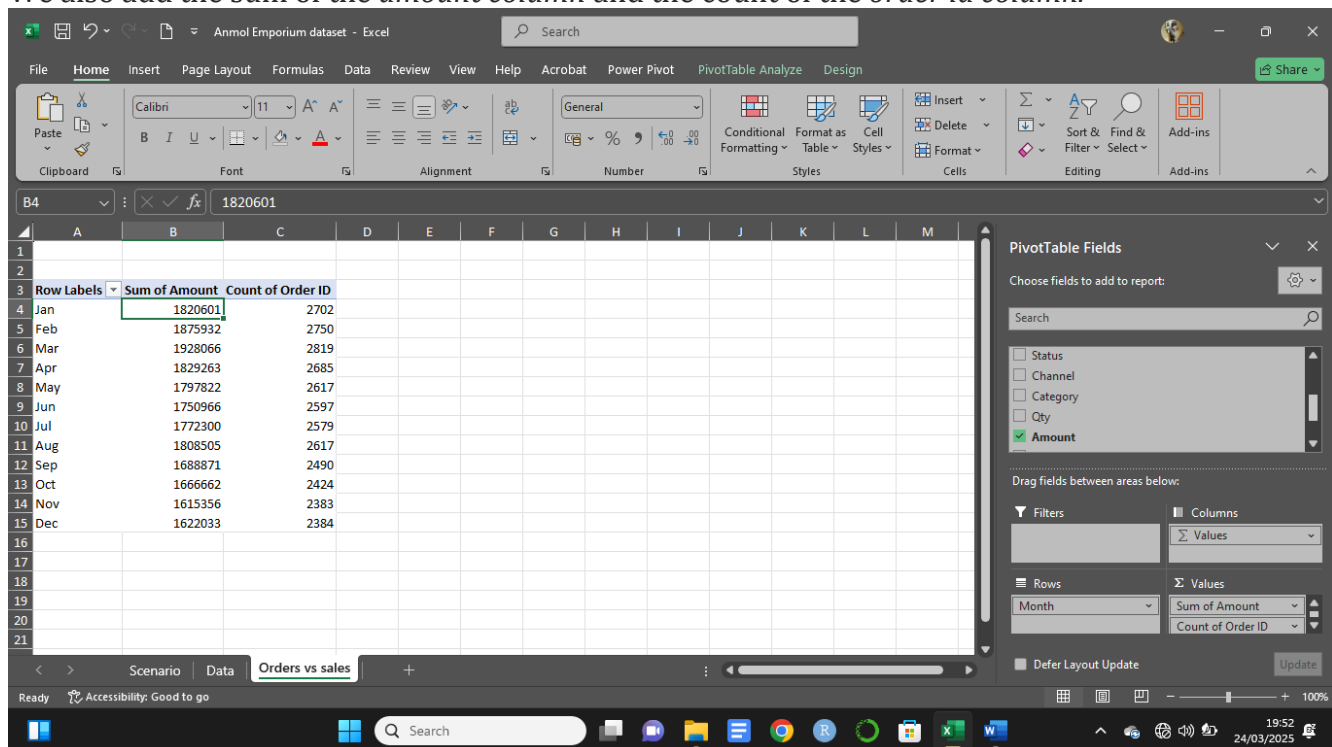
We have to create a Pivot Table for further analysis.

Analysis 1

First, create pivot Table sheet for the first and second business questions and named it — ‘sales by orders’.

Add month in the row section.

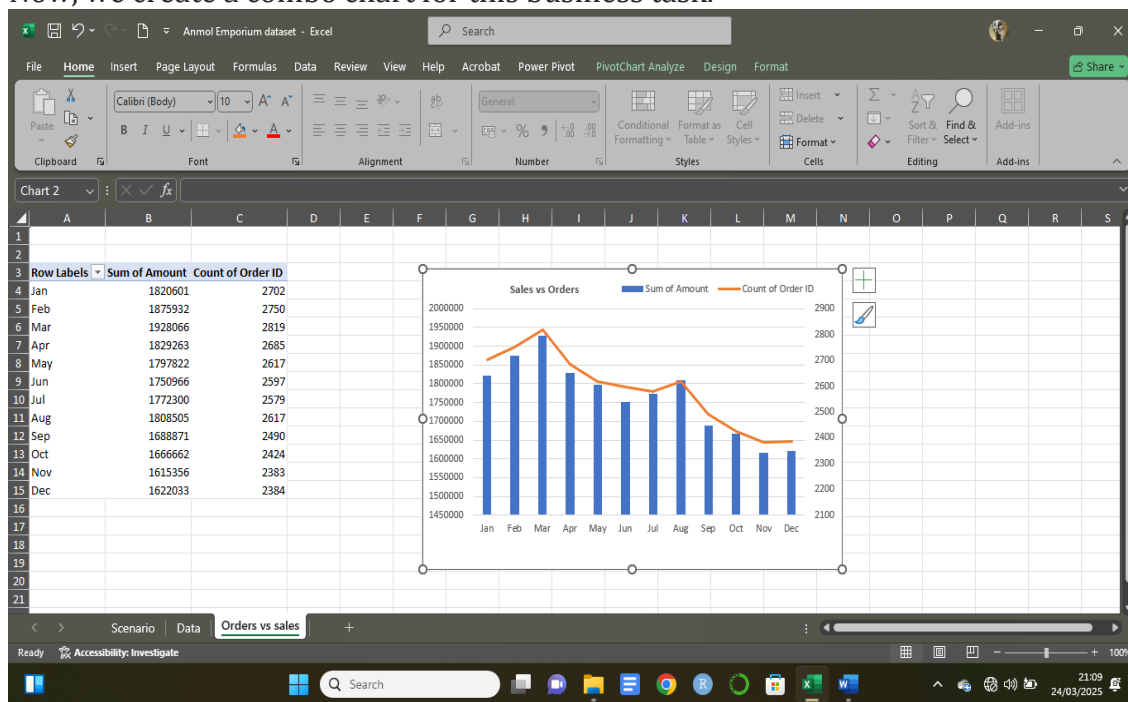
We also add the sum of the amount column and the count of the order id column.



Row Labels	Sum of Amount	Count of Order ID
Jan	1820601	2702
Feb	1875932	2750
Mar	1928066	2819
Apr	1829263	2685
May	1797822	2617
Jun	1750966	2597
Jul	1772300	2579
Aug	1808505	2617
Sep	1688871	2490
Oct	1666662	2424
Nov	1615356	2383
Dec	1622033	2384

Removed the grand total.

Now, we create a combo chart for this business task.

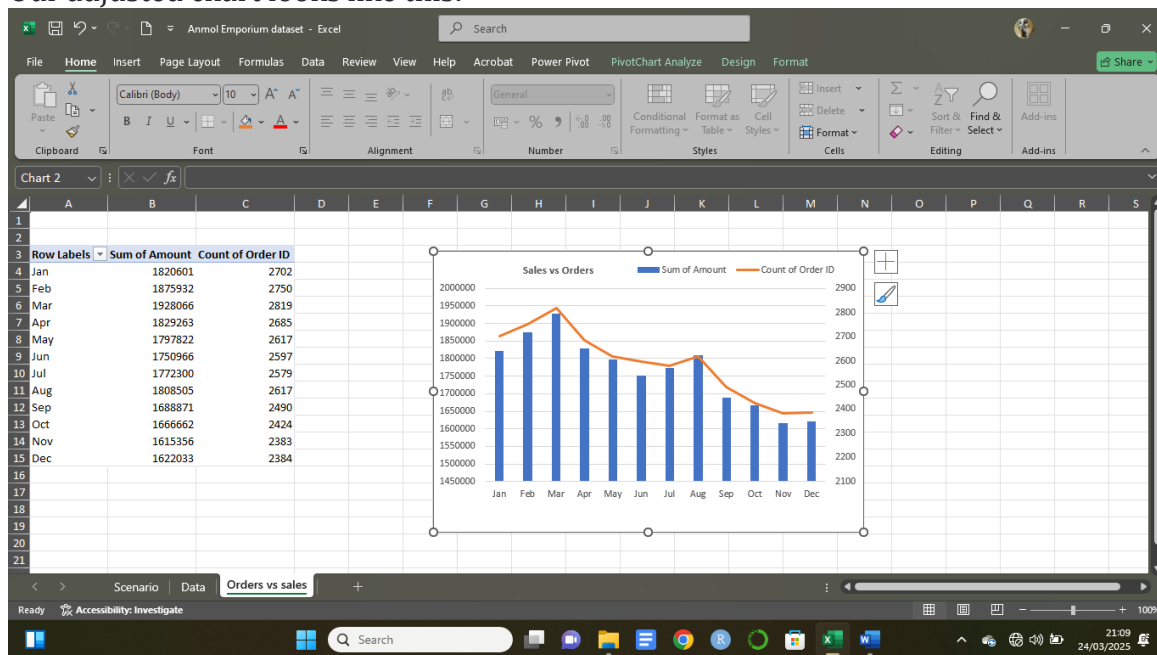


Combo Chart of sales vs orders

Let's customize this chart:

- Off the gridlines.
- Adjust the Chart Title and Legend by re-position them.
- Adjust the column axis (click on the column axis, then the axis option, then numbers, then format code and type **0.00,, "M"** to adjust it.

Our adjusted chart looks like this:



Formatted Chart

From that chart, we can clearly show the first two business questions i.e.,

- Compare the sales by orders using a single chart.

- Which month got the highest sales and orders?

The highest month of sales and orders is *March*.

Analysis 2

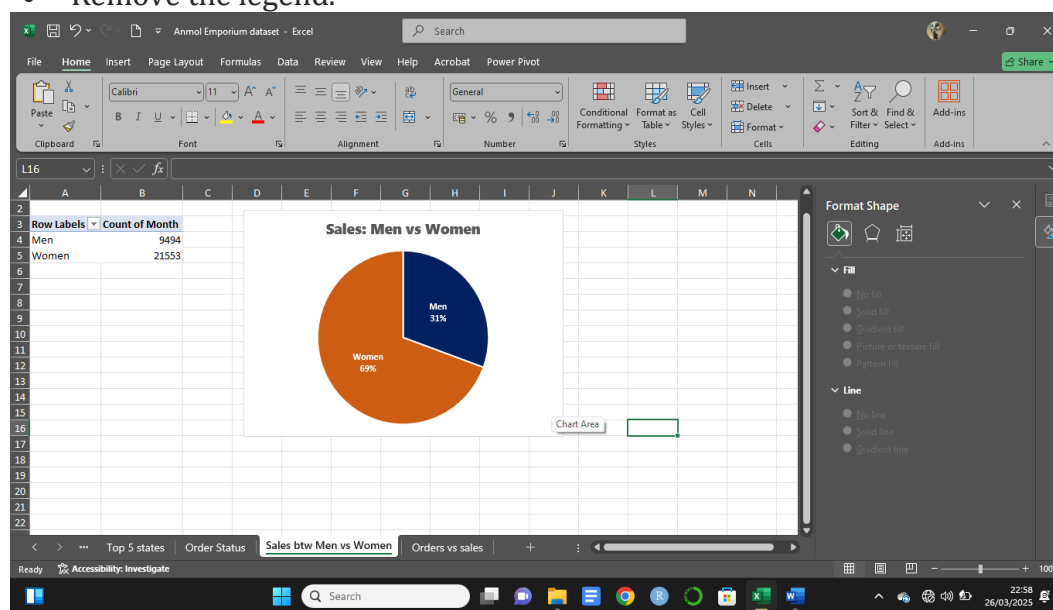
Now, we are creating visualization for the third business question i.e., Who purchased more — men or women in 2022?

We create another Pivot Table Sheet and name it — ‘Sales by **men & women**’.

We add a *gender column* in the *row section* and *amount column* in the *value section*.

Now, we create the pivot chart for this pivot table. Let’s customize it:

- first, we change the chart title name.
- Second, label the pie chart with the “data callout” option.
- Remove the legend.



In this chart, women purchase more than men.

Analysis 3

Now, we are creating the visualization for the fourth business question i.e., What is the different order status in 2022?

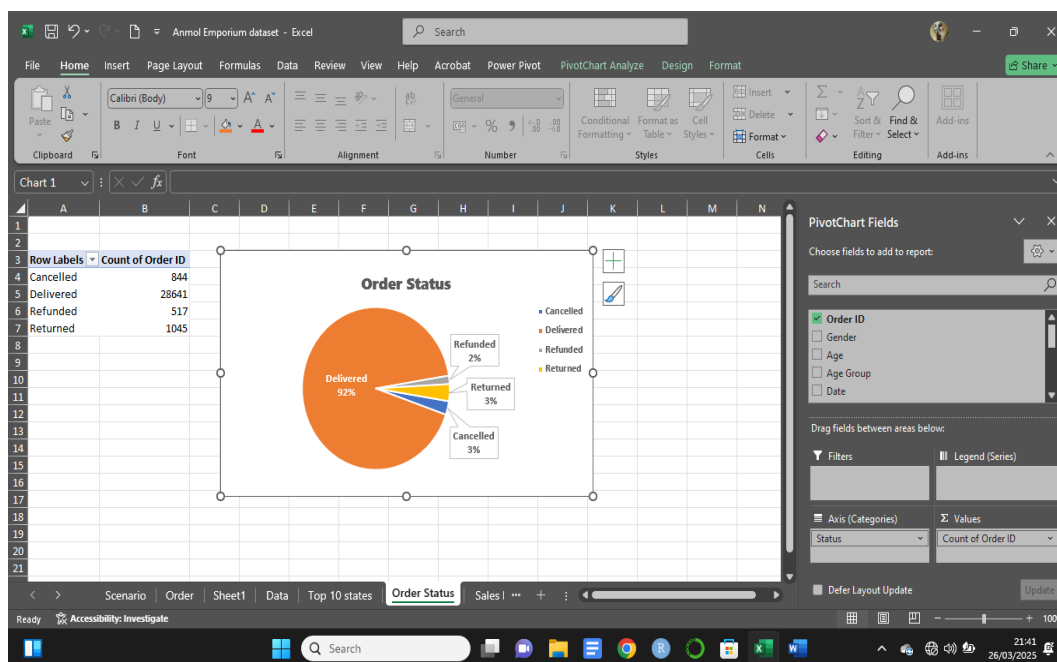
We create another Pivot Table Sheet and name it — ‘**Order Status**’.

We add a *status column* in the *row section*.

We add an *order id* column in the *value* section. Now, we create the pivot chart for this pivot table.

Let's customize it:

- First, Hide all the field buttons
- Second, label the pie chart with the “data callout” option.
- Remove the legend
- Rename the title to ‘Order Status’.



Analysis 4: List the top 5 states that contribute to the sales.

Create another Pivot Table Sheet and name it — ‘**Top 5 states.**’

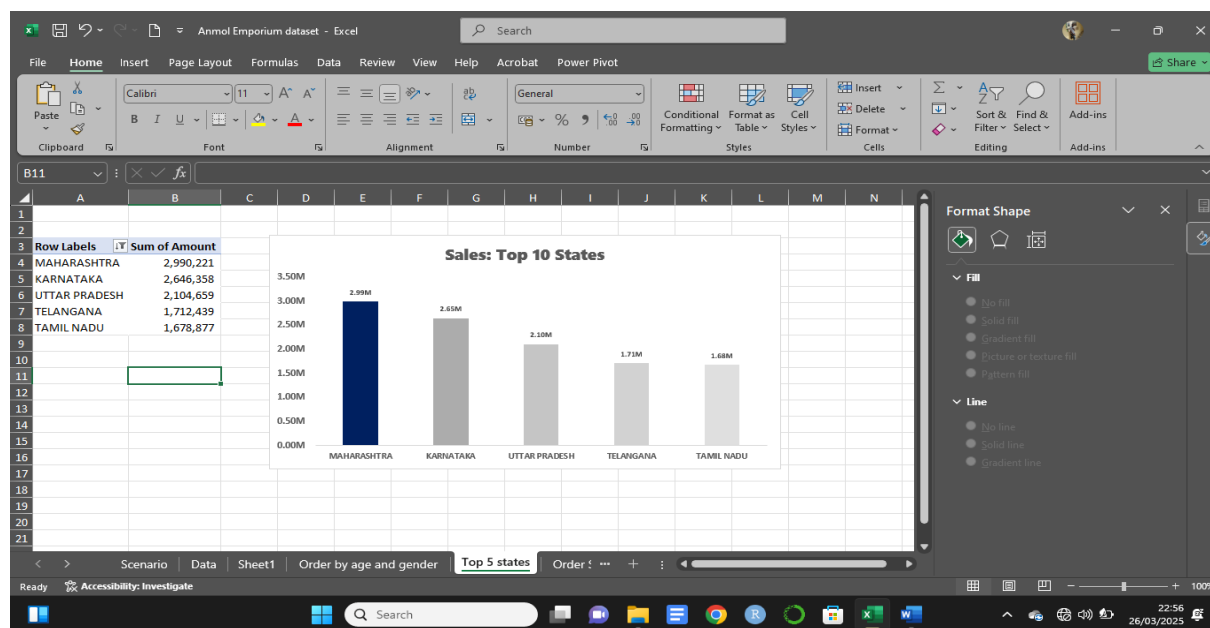
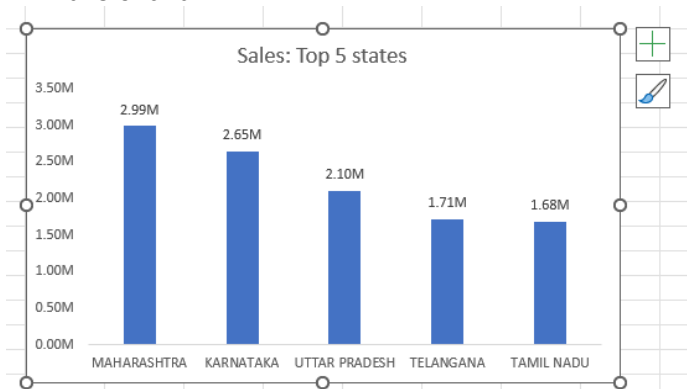
Add *status* column in the *row* section and *order id* column in the *value* section.

Now, we create the pivot chart for this pivot table.

Let's customize it:

- First, Hide all the field buttons
- Rename the title to ‘Sales: Top 5 states.’
- Remove the legend and gridlines from the chart.
- Adjust the column axis (click on the column axis, then the axis option, then numbers, then format code and type **0.00,”M”** to adjust it.

- Add the label outside the bars and adjust the label values in millions with the above method and format the chart.



The highest sales are in Maharashtra, Karnataka, Uttar Pradesh, Telangana, and Tamil Nadu.

Analysis 5

Now, we are creating visualization for the sixth business question i.e., Relationship btw age and gender based on number of orders.

We create another Pivot Table Sheet and name it — ‘Order by Age & Gender’.

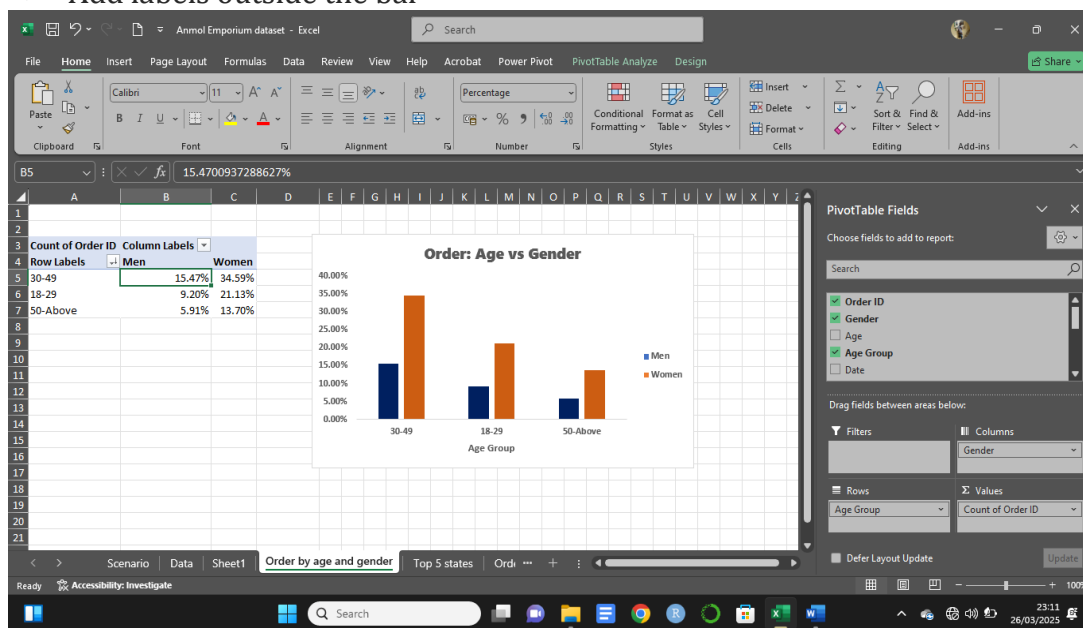
Add an *Age Group* column in the row section and *Gender* column in the column section. Then count of order *id* in the value section.

Convert this data in percent of grand total (right-click on the pivot table, then ‘show value as’ and then “% of grand total”.

Now, create the pivot chart for this pivot table and customize it:

- First, Hide all the field buttons
- Add chart title name — ‘Order: Age vs Gender’.

- Remove the gridlines.
- Adjust the legend
- Add labels outside the bar



Analysis 6

Now, we are creating the visualization for the seventh business question i.e., Which channel is contributing to maximum sales?

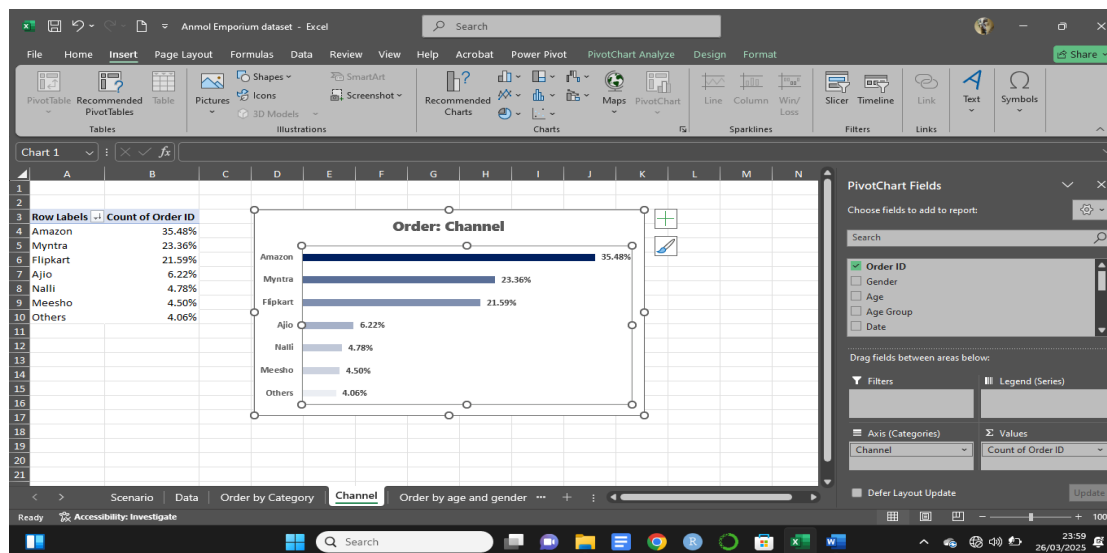
We create another Pivot Table Sheet and name it — ‘Channels’.

Add a *Channels* column in the row section and *count of order id* in the value section.

Convert this data in percent of grand total (right-click on the pivot table, then ‘show value as’ and then “% of grand total”.

Now, we create the pivot chart for this pivot table and customize it:

- First, Hide all the field buttons
- Add chart title name — ‘Order: Channels’
- Remove the legend, Add bar chart and label with the “outside end” option



Analysis 7

Now, we are creating the visualization for the eight-business question i.e., Highest Selling Category?

We create another Pivot Table Sheet and name it — ‘Highest selling category’.

Add a *Category* column in the row section and *count of order id* in the value section.

Convert this data in percent of grand total (right-click on the pivot table, then ‘show value as’ and then “% of grand total”.

Now, we create the pivot chart for this pivot table and customize it:

- First, Hide all the field buttons
- Add chart title name — ‘Order by Category’
- Remove the legend, Add bar chart and label with the “outside end” option

Dashboard

After all the analysis and visualization, we put all charts and graphs on one sheet.

We create a new sheet and build a dashboard on that sheet.

We put all the viz. on that sheet and adjust all of them.

Add Slicer

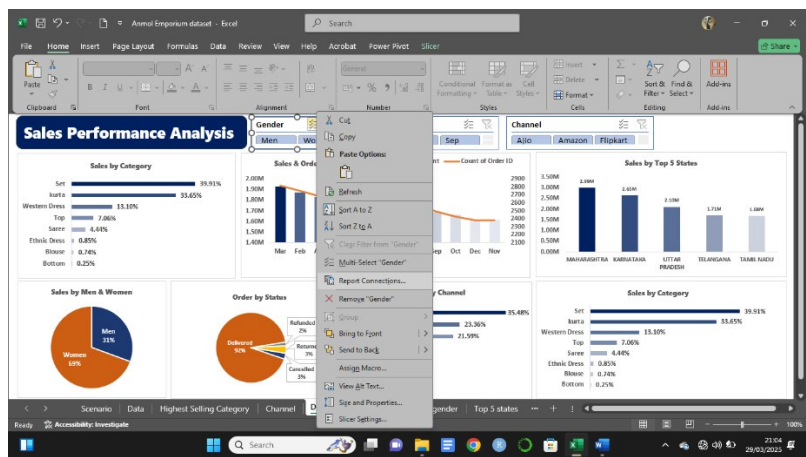
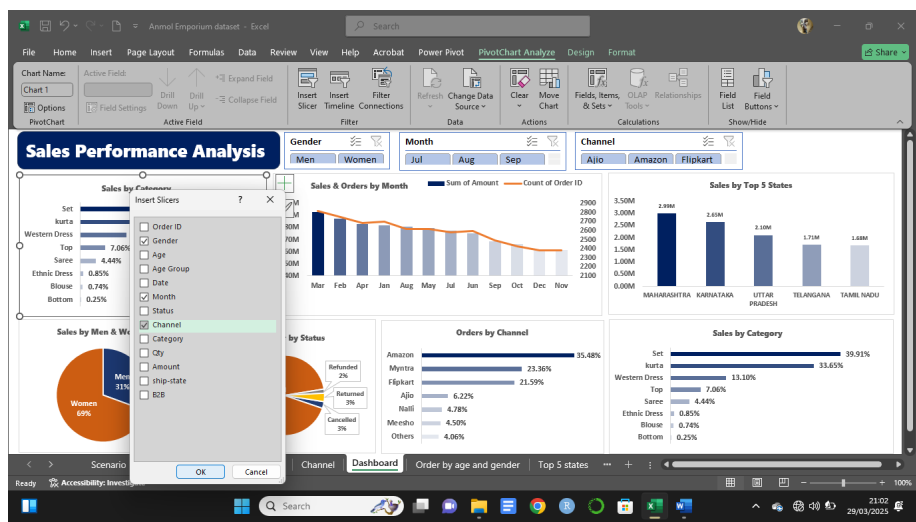
Now, we have to do one last thing i.e., *Add Slicer* to this dashboard.

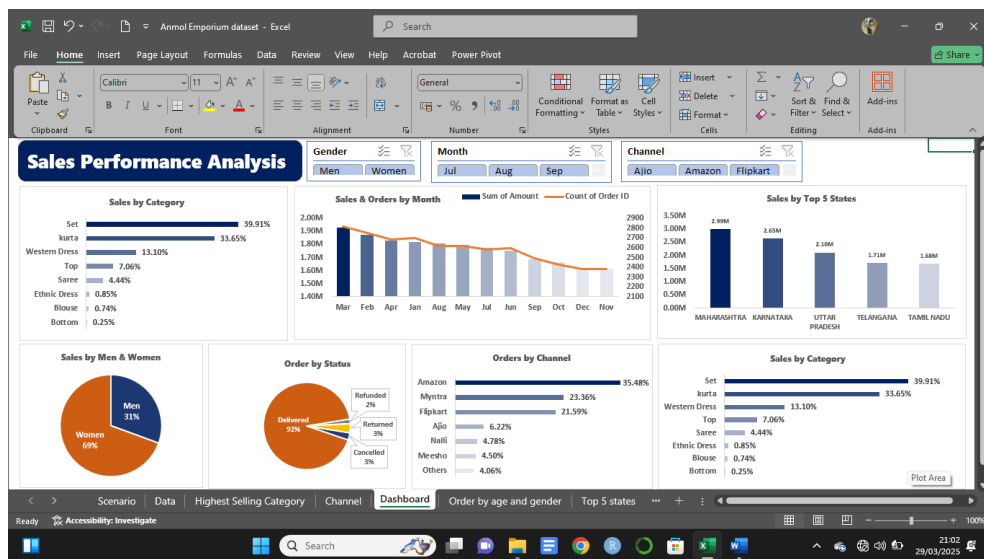
We add three slicers:

- Gender
- Month
- Channel

Adjust these slicers to the dashboard. Now, connect all these slicers to the visualization.

Right-Click on the slicer — select ‘report connections’ — Tick all the sections in this dialog box.





Insights and Recommendations

Detailed Insights:

- **Customer Gender:** Female customers account for **65%** of total purchases, indicating a significantly higher conversion rate compared to male shoppers.
- **Top Performing States:** **Maharashtra, Karnataka, and Uttar Pradesh** collectively contribute to **35%** of overall sales, making them the leading regions in terms of market demand.
- **Age Group Performance:** The **30–49 age group** (adults) drives the majority of sales, contributing to **50%**, suggesting strong purchasing power and engagement.
- **Product Categories:** Sales are primarily concentrated in the **Set, Kurta, and Western wear** categories, which together account for **85%** of total category sales.
- **Sales Channels:** **Amazon, Myntra, and Flipkart** dominate distribution, generating **80%** of total channel-based revenue.

Conclusion & Recommendations:

Focus marketing efforts on **female consumers aged 30–49** residing in **Maharashtra, Karnataka, and Uttar Pradesh**. Utilize high-performing channels—**Amazon, Myntra, and Flipkart**—to deliver targeted ads, promotional offers, and personalized coupons, particularly for top-selling product categories to maximize sales growth and ROI.